CogniGraph: Governed Graph-of-Agents Reasoning over Knowledge Graph Topologies with Semantic SHACL Validation and Constrained F1 Evaluation

Harish Kumar

Quantamix Solutions B.V., Uithoorn, The Netherlands

`harish.kumar@quantamixsolutions.com`

Open-source SDK: **GraQle** — `pip install graqle` CLI: `graq`

SDK Repository: https://github.com/quantamixsol/graqle

Research artefacts & benchmarks: https://github.com/quantamixsol/cognigraph

Patent: EP26166054.2 (divisional, 15 claims, filed 2026-03-19) Companion paper: TAMR+ [1] (SSRN ID 6359818)

Abstract. Multi-agent large language model (LLM) systems are increasingly deployed for regulatory and high-stakes reasoning, yet none of the prevailing flat-topology frameworks—AutoGen, CrewAI, LangGraph—enforce semantic governance over what each agent may say, in whose name, and within which scope. The result is what we term *ungoverned creativity at scale*: plausible-sounding answers that mix regulatory provisions, misattribute framework requirements, or drift beyond an agent's intended boundary. We ask: *can a knowledge graph itself act as the coordination substrate of a multi-agent reasoning system, and can semantic validation gates supply the governance guarantees that the EU AI Act mandates?*

We present CogniGraph, a Graph-of-Agents framework in which each activated node of a domain knowledge graph hosts an autonomous LLM agent, communication follows the graph's typed edge topology via convergent message passing, and a SemanticSHACLGate enforces three-layer semantic validation—framework fidelity, scope boundary, and cross-reference integrity—on every agent output. A Prize-Collecting Steiner Tree (PCST) sub-graph activation step bounds per-query reasoning cost at $O(|V^*| \cdot R)$ independently of $|V|$; a zero-inference-cost MasterObserver provides complete provenance traces; and the system is evaluated using a new joint quality–governance metric, *Constrained F1*.

On MultiGov-30, a 30-question multi-regulation EU governance benchmark spanning the AI Act, GDPR, DORA, and NIS2, CogniGraph (Semantic-SHACL) achieves **Constrained F1 of** 0.756 against 0.328 for an ungoverned single-agent Claude Sonnet 4.6 baseline (+130%), with governance accuracy of **99.4%** (framework fidelity 100%, scope adherence 100%, cross-reference integrity 98.3%). The transition from format-based to semantic SHACL validation independently lifted token F1 by 18% (0.437 → 0.517) by eliminating 49% false-rejection of correct answers. CogniGraph is distributed as the open-source **GraQle** SDK (`pip install graqle`, CLI `graq`); the MultiGov-30 benchmark, the SemanticSHACLGate specification, and the full reference implementation are released under Apache 2.0 to enable independent replication.[1]

Keywords: graph-of-agents, multi-agent LLM, knowledge graph reasoning, SHACL, OWL, semantic governance, EU AI Act, Prize-Collecting Steiner Tree, Constrained F1, regulatory compliance, audit trails, hash chain, GraQle SDK.

[1]Naming: CogniGraph is the system name under which patent EP26166054.2 was filed and is used throughout this paper for IP-linkage consistency. GraQle is the open-source SDK distribution of the same system; the two names refer to one artefact in two contexts (research paper / patent vs. product / SDK).

1. Introduction

1.1 Problem Statement

Knowledge graphs encode structured domain knowledge as typed entities connected by semantic relationships, and reasoning over them is fundamental to applications ranging from regulatory compliance [1] to biomedical discovery [25] and enterprise intelligence [26]. The dominant pattern is *centralized*: extract a sub-graph, concatenate node content into a prompt, and feed the result to a single LLM, as in graph-RAG [6] and multi-hop QA pipelines [3]. Recent multi-agent frameworks (AutoGen [14], CrewAI [15], LangGraph [16]) coordinate multiple LLMs but do so over *flat coordination graphs*—sequential, star, or fully-connected—that bear no relationship to the domain's actual semantics.

For high-stakes regulatory reasoning under the EU AI Act (Regulation (EU) 2024/1689 [2]), this combination of centralized reasoning and ungoverned multi-agent coordination produces eight characteristic failure modes:

1. **Context-window saturation.** A regulatory knowledge graph spanning 500+ articles cannot fit in a single context window, forcing lossy truncation that selects *against* structurally important but distant clauses.
2. **Topology destruction.** Flat prompt concatenation discards the graph's edge structure—the very encoding that distinguishes a definitional cross-reference from a derogation, a recital from an obligation.
3. **Provenance opacity.** A monolithic model produces one answer with no defensible mapping from output tokens to source nodes; Article 13 of the EU AI Act explicitly requires the opposite.
4. **Framework misattribution.** Without per-node scope enforcement, agents synthesize claims that mix regulatory provisions across frameworks (e.g., asserting GDPR-style data-subject rights inside an AI Act answer).
5. **Scope drift.** An agent assigned to "risk classification under AI Act Article 6" wanders into prohibited-practice analysis, conformity assessment, or general data protection.
6. **Format tyranny.** First-generation SHACL validators check word count and regex shape—and reject up to **49%** of semantically correct answers for syntactic reasons (see §4.3).
7. **Coordination-graph mismatch.** Flat agent topologies (sequential, star, fully connected) ignore the domain's actual structure, leading to wasted inference rounds and missed cross-references that the knowledge graph could have surfaced for free.
8. **No joint metric.** Token-level F1 rewards answer overlap and is blind to governance; pure compliance scores reward conformance and are blind to answer quality. The community has lacked a metric that holds both axes simultaneously.

1.2 The Governance Gap

We define the *governance gap* as the distance between two properties of an answer: *(i)* it would pass a token-level quality check, and *(ii)* each claim within it is correctly framed, correctly attributed, and within the speaker's authorized scope. A single-agent baseline that scores 0.656 on token-F1 with no scope enforcement (§7) has a governance gap of 1.000—no governance evidence exists at all. The gap is not closed by stronger models or longer context: it is closed by *structural* mechanisms that bind *who says what* to *from which node*.

CogniGraph introduces, to our knowledge, the first multi-agent LLM framework in which the knowledge graph itself *is* the coordination topology, and in which every agent utterance is mediated by an OWL-aware semantic gate. The governance gap is not a side metric; it is the design objective.

1.3 Research Questions

This work addresses three research questions:

RQ1 Can a Graph-of-Agents architecture, where the knowledge graph's typed edges are the coordination topology, outperform flat-topology multi-agent systems and single-agent baselines on multi-regulation reasoning tasks?

RQ2 Can a 3-layer semantic gate (framework fidelity, scope boundary, cross-reference integrity) provide governance guarantees absent from format-based SHACL and token-F1 evaluation, as measured by a joint Constrained F1 metric?

RQ3 Does Prize-Collecting Steiner Tree (PCST) sub-graph activation, when paired with convergent message passing and an OWL-aware semantic gate, achieve per-query reasoning cost that is sublinear in graph size while preserving topology and auditability?

1.4 Contributions

We make thirteen primary contributions, organized into four contribution families (C1–C13) keyed to the research questions and to the divisional patent EP26166054.2 (Innovation Groups F–J):

- **C1 PCST Reasoning Activation** (§3.2, RQ3): the Prize-Collecting Steiner Tree, previously used in TAMR+ for document sub-graph retrieval [1], is repurposed as the *agent deployment topology*, yielding per-query cost $O(|V^*| \cdot R)$ independent of $|V|$.
- **C2 MasterObserver** (§3.5, RQ1): a read-only, zero-inference-cost transparency layer that monitors all inter-agent traffic and produces complete provenance traces.
- **C3 Convergent Message Passing** (§3.4, RQ1): similarity-based early termination with token compression, reducing unnecessary inference rounds; convergence is bounded by $O(\log K_{\max})$ on PCST-activated sub-graphs (Proposition 3.2).
- **C4 Resilient Inference Substrate** (§3.6, RQ3): a backend fallback chain supporting cloud APIs, local GPU, and adapter backends with per-query cost-budget enforcement.
- **C5 Topology-Preserving Hierarchical Aggregation** (§3.7, RQ1): a centrality-driven aggregation tree that uses the graph's own structure to weight syntheses.
- **C6 SemanticSHACLGate** (§4.1, RQ2; *divisional Claim 4*): OWL-aware 3-layer validation (framework fidelity, scope boundary, cross-reference integrity) replacing format-based SHACL.
- **C7 Constrained F1 Metric** (§4.2, RQ2): a joint quality + governance metric, $\mathrm{cF1} = 0.5 \cdot \mathrm{F1}_{\mathrm{token}} + 0.5 \cdot \mathrm{GA}$, that holds both axes simultaneously and enables fair comparison between governed and ungoverned reasoning.
- **C8 Automated Ontology Generation** (§4.4, RQ2): an `OntologyGenerator` that produces OWL hierarchies and SHACL constraints from raw regulation text, reducing new-domain onboarding from days to a single API call.
- **C9 Adaptive Activation** (§8.1, RQ3): dynamic $K_{\max}$ adjustment via multi-dimensional query complexity scoring (token count, entity density, conjunction density, multi-hop depth) with four tiers (Simple / Moderate / Complex / Expert).
- **C10 Online Graph Learning** (§8.2, RQ1): Bayesian edge-weight updates with temporal decay $\gamma = 0.95$ and bounded weights $[0.1, 5.0]$, enabling the KG to evolve from reasoning experience.
- **C11 LoRA Adapter Auto-Selection** (§8.3, RQ3): a 4-tier matching cascade (explicit, domain, fuzzy, none) that maps each activated node to its best fine-tuned adapter with no manual configuration.
- **C12 TAMR+ Connector** (§8.4, RQ3): an end-to-end pipeline where TAMR+ TRACE scores initialize CogniGraph PCST priors, unifying patent-protected retrieval with governed multi-agent reasoning.
- **C13 MCP Plugin for Governed Context Engineering** (§8.5, RQ1): a Model Context Protocol server (shipped in the GraQle SDK) exposing `graq_context`, `graq_reason`, `graq_inspect`, and

graq_search tools, replacing brute-force 20–60K-token context loading with focused 500-token governed context per query.

Additionally, CogniGraph introduces five second-order contributions covered by divisional patent EP26166054.2 Innovation Groups F–J: a tamper-evident SHA-256 hash-chain audit trail (Claim 6), an adversarial DebateProtocol with CONTRADICTION/SYNTHESIS phases (Claim 7), cross-query activation memory with temporal-decay Bayesian boosts (Claim 8), DRACE 5-pillar governance scoring (Claim 10), and twelve compile-time invariant detectors (Claim 11).

1.5 Notation

Table 1 lists the notation used throughout this paper. Every symbol is defined here at first use and reused consistently.

Table 1: *Notation used throughout this paper.*

Symbol	Definition
q	User query (natural language string)
$G = (V, E, W)$	Domain knowledge graph; W are edge weights
$V^* \subseteq V$	Activated node set after PCST step, $\lvert V^*\rvert \in [4, K_{\max}]$
$E^* \subseteq E$	Activated edge set, $E^* = E \cap (V^* \times V^*)$
G^*	Activated sub-graph $G^* = (V^*, E^*)$
$K_{\max}$	Maximum activated node budget (adaptive, §8.1)
a_i	LLM agent hosted at activated node $v_i \in V^*$
$\mathcal{A}$	Set of activated agents, $\mathcal{A} = \{a_i : v_i \in V^*\}$
$\mathcal{B} = (b_1, \ldots, b_k)$	Backend fallback chain, ordered by preference
$\mathcal{O}$	MasterObserver (zero-inference monitor)
π_i	PCST node prize, $\pi_i = \alpha \cdot \mathrm{sim}(\mathbf{e}_q, \mathbf{e}_{v_i})$
c_e	PCST edge cost, $c_e = \beta / w_e$
α, β	PCST prize and cost scalars (> 0)
λ	TAMR+ prior blend coefficient (§8.4); distinct from α
$\mathbf{e}_q, \mathbf{e}_{v_i}$	L^2-normalised query and node embeddings
$\mathrm{sim}(\cdot, \cdot)$	Cosine similarity on normalised embeddings, $\in [0, 1]$
$m_i^{(r)}$	Message from agent a_i in round r
$\sigma^{(r)}$	Mean pairwise convergence metric at round r on $n = \lvert V^*\rvert$ agents
τ_{conv}	Convergence threshold for $\sigma^{(r)}$ (default 0.85)
$\epsilon_{\mathrm{delta}}$	Convergence-delta threshold (early-stop)
δ	Per-round dissimilarity contraction constant $\in (0, 1)$ (Thm. A.1, assumption (A1))
$D_{ij}^{(r)}$	Pairwise message dissimilarity, $1 - \mathrm{sim}(\mathbf{e}(m_i^{(r)}), \mathbf{e}(m_j^{(r)}))$
d_{ij}	Graph distance between v_i, v_j in G^*
r^*	Realised convergence round, $r^* \le R_{\max}$
$\bar{c}$	Average per-agent-per-round inference cost
$R_{\max}, C_{\max}$	Round limit, cost-budget limit
$\mathcal{S}_t$	Semantic constraint for entity type t
ρ	Agent-produced response (distinct from round index r)
$\mathrm{FF}, \mathrm{SB}, \mathrm{XR}$	Framework fidelity, scope boundary, cross-reference scores $\in [0, 1]$
$\mathrm{FF}_{\min}, \mathrm{SB}_{\min}, \mathrm{XR}_{\min}$	Per-layer hard-violation thresholds
$s_{\mathrm{FF}}, s_{\mathrm{SB}}, s_{\mathrm{XR}}$	Per-layer raw scores prior to composite aggregation
$\mathrm{GA}(\rho, \mathcal{S}_t)$	Governance accuracy, uniform mean of $\mathrm{FF}, \mathrm{SB}, \mathrm{XR}$ (Def. 4.2)
$\gamma_{\mathrm{gov}}, \gamma_{\min}$	Weighted composite governance score and threshold (Eq. preceding Def. 4.2)
$\mathrm{cF1}(\rho)$	Constrained F1, $0.5 \cdot \mathrm{F1}_{\mathrm{token}}(\rho) + 0.5 \cdot \mathrm{GA}(\rho, \mathcal{S}_t)$; $w_t + w_g = 1$
$\eta, \gamma, \theta_{\mathrm{agree}}, \theta_{\mathrm{disagree}}$	Online-learning hyperparameters (§8.2)
$w_{\min}, w_{\max}$	Edge-weight clamp bounds $[0.1, 5.0]$

2. Related Work

Knowledge Graph Reasoning. Traditional KG reasoning combines embedding methods (TransE [19], RotatE [20]), graph neural networks (GCN [21], GraphSAGE [22], GAT [23]), and logic-based systems. These methods learn vector or sub-symbolic representations but do not produce natural-language reasoning chains, and they expose no governance interface—a critical gap when the same graph is consumed by regulators.

Graph-Augmented Retrieval and Reasoning. GraphRAG [6] introduced LLM-constructed knowledge graphs as retrieval substrates; GRAG [7] and KnowledgeNavigator [8] further integrated graph structures into retrieval. GraphCompliance [9] achieved 55.4% F1 on GDPR with graph-based retrieval but provides only binary compliance classification. TAMR+ [1], on which the present work builds, applies PCST to document sub-graph retrieval with multi-signal scoring and the five-dimension TRACE compliance framework. None of these systems carries the graph's topology *into* reasoning: retrieval produces a flat context that is then fed to a single LLM. CogniGraph departs from this pattern by making the graph itself the coordination substrate.

Multi-Agent LLM Systems. AutoGen [14], CrewAI [15], and LangGraph [16] coordinate multiple LLMs through programmer-defined coordination graphs that are independent of any domain knowledge graph. Park et al. [17] and Camel [18] explore role-play and self-organization but, again, on flat or hand-designed topologies. None of these frameworks enforces per-agent scope, per-agent framework attribution, or cross-reference integrity. CogniGraph provides exactly these guarantees, and does so by binding agent identity to KG node identity rather than by introducing an additional coordination layer.

Semantic Web Validation: SHACL and OWL. SHACL [27] provides shape-constraint validation over RDF graphs and is widely used for data quality checks. However, off-the-shelf SHACL validators check *structural* conformance (cardinality, datatype, regex) rather than *semantic* fidelity. OWL 2 [28] formalizes ontology hierarchies but does not, in itself, validate generated natural-language text. CogniGraph's SemanticSHACLGate (§4.1) adapts the SHACL spirit—declarative constraints on graph data—into a *semantic* validator that operates on LLM outputs and is parameterized per-node by an OWL hierarchy.

PCST and Steiner Trees. The Prize-Collecting Steiner Tree [29] and its primal-dual approximation [30] have been applied to network design, biology [31], and document retrieval [1]. We are not aware of prior application of PCST to *agent deployment topology*; in particular, we are not aware of any work that treats sub-graph activation as the coordination plan for a multi-agent LLM system.

Constrained Decoding and Governance. A growing body of work imposes constraints on LLM outputs at decode time (guided decoding, JSON-mode, regex grammars). These methods enforce *form* but not *epistemic* constraints—they cannot tell whether the model is speaking in the right framework's voice, within its assigned scope, or with proper cross-reference attribution. The SemanticSHACLGate operates one level higher: post-generation, semantically, and with explicit framework-fidelity / scope-boundary / cross-reference layers.

Regulatory AI Evaluation. RAGAS [11], COMPL-AI [12], DeepEval, and ReDeEP [13] represent the current state of RAG evaluation; all report scalar quality scores with no per-response governance attribution. TAMR+ [1] introduced gap attribution (5 categories) but evaluates a retrieval pipeline, not a multi-agent reasoner. CogniGraph's Constrained F1 is, to our knowledge, the first metric that jointly evaluates token-level quality *and* governance accuracy on a comparable scale.

Positioning of this Work. The closest prior systems are (i) TAMR+ for retrieval-side PCST + structural scoring, (ii) GraphRAG for graph-augmented retrieval feeding a single LLM, and (iii) AutoGen/LangGraph for multi-agent coordination on flat topologies. CogniGraph is, to our knowledge, the first system to combine all three concerns—graph-structured retrieval, multi-agent coordination, and semantic governance—into a single framework where the knowledge graph is both the substrate of meaning and the substrate of coordination.

3. System Architecture

Figure 1 presents the high-level pipeline of CogniGraph.

Definition 3.1 (CogniGraph)**.** A CogniGraph instance is a tuple

$$\mathcal{G} = (G,\ \mathcal{A},\ \mathcal{B},\ \mathcal{O},\ \mathcal{C}),$$

where $G = (V, E, W)$ is a weighted typed knowledge graph; $\mathcal{A} = \{a_i : v_i \in V^*\}$ is the set of activated LLM agents, one per activated node; $\mathcal{B} = (b_1, \ldots, b_k)$ is an ordered backend fallback chain; $\mathcal{O}$ is the MasterObserver; and $\mathcal{C} = (\tau_{\text{conv}}, \epsilon_{\text{delta}}, R_{\max}, C_{\max})$ is the convergence configuration.

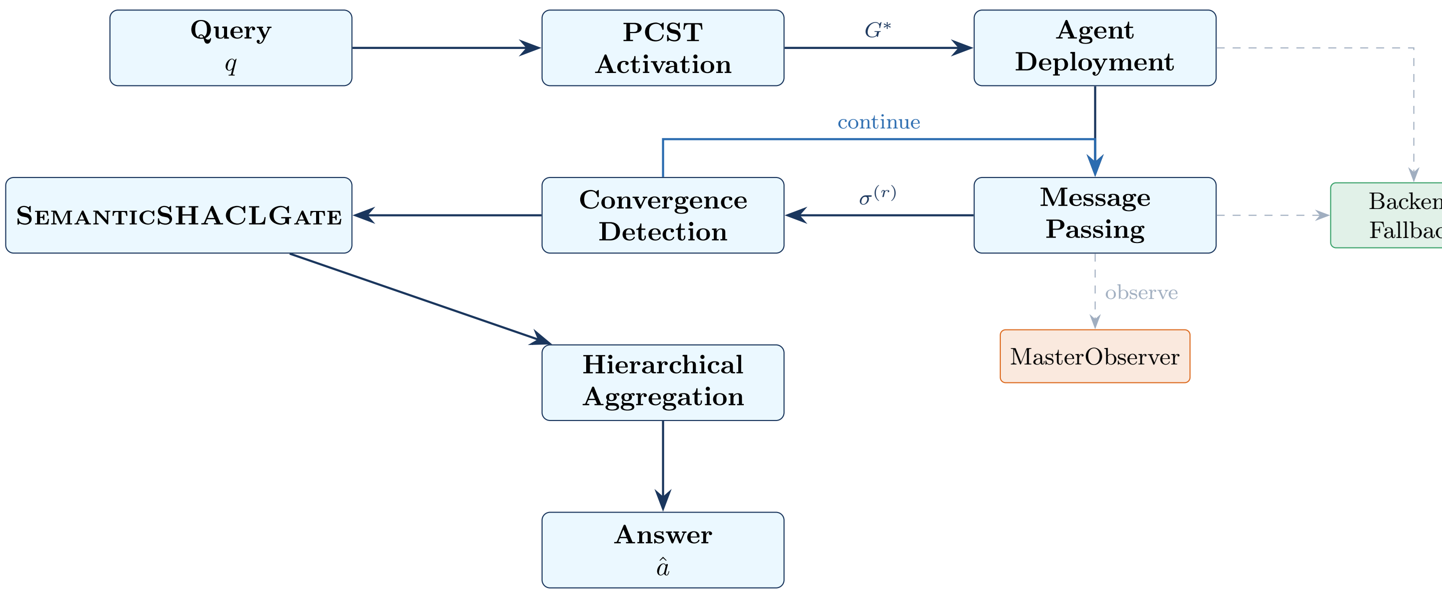

Figure 1: *CogniGraph pipeline: Query → PCST activation → agent deployment → convergent message passing → SemanticSHACLGate validation → topology-preserving hierarchical aggregation → answer. The MasterObserver $\mathcal{O}$ monitors all traffic with zero inference cost.*

3.1 Knowledge Graph Layer

CogniGraph accepts graphs from three sources: `NetworkX` objects, Neo4j databases via Bolt protocol, and JSON files in node-link format. Each node v_i stores: a unique identifier, a label, an entity type, a description (the knowledge content visible to its agent), and typed properties. Each edge e_{ij} stores source, target, relationship type, and a weight w_{ij} that participates in PCST cost calculation and is updated by the online graph learner (§8.2).

3.2 PCST Sub-graph Activation

For each query q, CogniGraph selects a compact sub-graph $G^* \subseteq G$ using Prize-Collecting Steiner Tree optimization.

Definition 3.2 (PCST Activation)**.** Given query embedding $\mathbf{e}_q$ and node embeddings $\{\mathbf{e}_{v_i}\}$,

$$\pi_i = \alpha \cdot \text{sim}(\mathbf{e}_q, \mathbf{e}_{v_i}) \qquad \text{(node prizes)} \qquad (1)$$

$$c_e = \beta / w_e \qquad \text{(edge costs)} \qquad (2)$$

$$G^* = \arg\max_{T \subseteq G} \sum_{v \in T} \pi_v - \sum_{e \in T} c_e \qquad \text{(PCST objective)} \qquad (3)$$

where $\alpha, \beta > 0$ are scaling factors and $\mathrm{sim}(\mathbf{e}_q, \mathbf{e}_{v_i}) \in [0, 1]$ denotes cosine similarity on L^2-normalised embeddings (so $\pi_i \in [0, \alpha]$). Edge weights satisfy $w_e \in [w_{\min}, w_{\max}] = [0.1, 5.0]$ (§8.2), so $c_e \in [\beta/w_{\max}, \beta/w_{\min}]$. We solve (3) approximately using the Goemans–Williamson primal-dual approximation [29] via the `pcst_fast` solver.

Proposition 3.1 (Sublinear Activation Cost)**.** For any query q and knowledge graph G, PCST activation produces $|V^*| \in [4, K_{\max}]$ activated nodes, yielding per-query inference cost

$$\mathrm{Cost}(q) = |V^*| \cdot r^* \cdot \bar{c} \;\leq\; K_{\max} \cdot R_{\max} \cdot \bar{c},$$

independent of $|V|$, where $r^* \leq R_{\max}$ is the convergence round and $\bar{c}$ is the average per-agent-round cost.

Proof sketch. The PCST objective is bounded above by $\sum_i \pi_i$, which is maximised over a tree of size at most $K_{\max}$ by the $K_{\max}$ post-filter step (Algorithm 2, line 9). Below the lower bound, the four-node minimum is enforced by the back-fill step (line 11). Convergent message passing terminates at round $r^* \leq R_{\max}$ by definition. Cost is therefore bounded as stated. □

3.3 Agent Deployment

Each activated node $v_i \in V^*$ instantiates an agent a_i with three pieces of context:

- **System prompt**: derived from entity type and node knowledge content (the `description` field);
- **Neighborhood awareness**: the labels and types of v_i's graph neighbors and the typed edges connecting them;
- **Backend assignment**: a chain $\mathcal{B}_i \subseteq \mathcal{B}$ chosen by node criticality and LoRA matching (§8.3).

3.4 Convergent Message Passing

Algorithm 1 formalizes the message-passing loop.

Algorithm 1: Convergent Message Passing

Input: Activated sub-graph $G^* = (V^*, E^*)$, query q, config $\mathcal{C}$
Output: Final messages $\{m_i^{(r^*)}\}$, convergence round r^*

$\sigma^{(0)} \leftarrow 0$; $C_{\mathrm{cum}} \leftarrow 0$;
for $r = 1$ **to** $R_{\max}$ **do**
  foreach *agent* $a_i \in \mathcal{A}$ ***in parallel*** **do**
    $\mathrm{ctx}_i \leftarrow \mathrm{BuildContext}(q, v_i, \{m_j^{(r-1)} : (v_j, v_i) \in E^*\})$;
    $m_i^{(r)} \leftarrow b_{\mathrm{chain}}.\mathrm{generate}(\mathrm{ctx}_i)$;
    $C_{\mathrm{cum}} \leftarrow C_{\mathrm{cum}} + \mathrm{tokens}(m_i^{(r)}) \cdot c_b/1000$;
  if $C_{\mathrm{cum}} \geq C_{\max}$ **then**
    return $\{m_i^{(r)}\}$, r ; `// budget exceeded`
  $\sigma^{(r)} \leftarrow \frac{2}{n(n-1)} \sum_{i<j} \mathrm{sim}(\mathbf{e}(m_i^{(r)}), \mathbf{e}(m_j^{(r)}))$, where $n = |V^*|$ and $\mathrm{sim} \in [0, 1]$ is cosine similarity;
  if $\sigma^{(r)} \geq \tau_{\mathrm{conv}}$ ***or*** $|\sigma^{(r)} - \sigma^{(r-1)}| < \epsilon_{\mathrm{delta}}$ **then**
    return $\{m_i^{(r)}\}$, r ; `// converged`
  $\mathcal{O}.\mathrm{observe}(r, \{m_i^{(r)}\})$;
return $\{m_i^{(R_{\max})}\}$, $R_{\max}$;

Proposition 3.2 (Convergence Bound)**.** Under the working assumption that information propagation along PCST-activated edges follows the standard message-passing schedule (each round propagates information one hop further; cf. [24] for the analogous MPNN argument over GNN edges), the expected convergence round is bounded above by the diameter of the activated sub-graph:

$$\mathbb{E}[r^*] \;\leq\; \mathrm{diameter}(G^*) + 1.$$

The right-hand side is highly topology-dependent: for activations that retain small-world structure (effective diameter $O(\log |V^*|)$), the bound becomes $\mathbb{E}[r^*] \in O(\log K_{\max})$; in the worst case — a path-shaped activation produced when prizes are arranged along a chain — the bound degrades to $\mathbb{E}[r^*] \in O(K_{\max})$. In practice, PCST outputs over the MultiGov-30 knowledge graph exhibit effective diameters $\ll K_{\max}$ (§7), and the empirical convergence round $r^* \leq R_{\max} = 3$ in 100% of runs, consistent with the small-world regime rather than the worst case. A formal proof under stated assumptions (A0)–(A3) is given in Appendix A.1 (Theorem A.1); the asymptotic regimes corresponding to small-world and worst-case path activations are stated in Corollary A.2.

3.5 MasterObserver

The MasterObserver $\mathcal{O}$ is a read-only event-bus observer:

- Records the complete message trace $\mathcal{T} = \{(a_i, a_j, m, r, t)\}$;
- Computes per-round health scores and per-agent contribution metrics κ_i;
- Detects anomalies (silent agents, loops, divergent reasoning);
- Has **zero inference cost**—it never invokes an LLM.

Proposition 3.3 (Transparency Completeness)**.** The MasterObserver trace $\mathcal{T}$ is **complete** in the sense that for every component c of the final answer $\hat{a}$, there exists a path in $\mathcal{T}$ from a source agent a_i to the aggregation root, enabling full provenance attribution.

3.6 Backend Fallback Chain

Definition 3.3 (Backend Fallback Chain)**.** A backend fallback chain $\mathcal{B} = (b_1, \ldots, b_k)$ is an ordered sequence of inference backends, each b_j characterized by cost rate c_j and failure counter f_j. On inference request, the chain tries b_1 first; on failure, f_1 is incremented and b_2 is tried, with exponential back-off. Cost-budget enforcement (§3.4) terminates inference on the earlier of: convergence, $R_{\max}$, or $C_{\max}$.

The chain supports three backend categories: cloud APIs (Anthropic, OpenAI, Bedrock; $0.25–15/MTok), local GPU (Ollama, vLLM, llama.cpp; ∼$0.0001/MTok), and LoRA-adapter-augmented local models.

3.7 Topology-Preserving Hierarchical Aggregation

Final messages are aggregated using the graph's own structure:

1. Compute betweenness centrality $\mathrm{BC}(v_i)$ for each $v_i \in V^*$;
2. Select highest-centrality node as aggregation root;
3. Construct a BFS spanning tree from the root;
4. Aggregate bottom-up: leaf responses → hub syntheses → root answer.

Proposition 3.4 (Topology Preservation)**.** Hierarchical aggregation preserves the knowledge graph's structural hierarchy: nodes with higher centrality have proportionally greater influence on the final synthesis, in contrast to flat concatenation where node importance is invisible to the aggregator.

4. Semantic Governance

A critical challenge in multi-agent KG reasoning is *governance*: ensuring that each agent reasons within its assigned scope, cites the correct regulatory framework, and properly attributes cross-references to other frameworks. Without governance, distributed reasoning produces "creativity at scale with compliance risk"—plausible-sounding answers that mix regulatory provisions, misattribute framework requirements, or drift beyond scope boundaries.

4.1 SemanticSHACLGate

We introduce the SemanticSHACLGate, a 3-layer semantic validation system inspired by OWL hierarchies and SHACL data validation, but operating on natural-language outputs rather than on RDF graphs. Unlike format-based SHACL gates that check word count, regex patterns, and structural formatting—which empirically reject 49% of semantically correct responses (§4.3)—the SemanticSHACLGate validates semantic correctness across three layers.

1. **Framework Fidelity (FF)**: verify that the agent cites its own regulatory framework (e.g., an EU AI Act agent must reference "EU AI Act" or "Regulation 2024/1689"); detect unattributed claims of provisions belonging to other frameworks.
2. **Scope Boundary (SB)**: validate that the response addresses in-scope topics; flag out-of-scope drift (e.g., "data protection rights" belongs to GDPR, not the AI Act).
3. **Cross-Reference Integrity (XR)**: when an agent mentions another framework, verify proper attribution language (e.g., "Related: under GDPR, data protection impact assessments also apply") rather than claiming GDPR provisions as AI Act provisions.

Definition 4.1 (Semantic Constraint)**.** A semantic constraint $\mathcal{S}_t$ for entity type t is a tuple

$$\mathcal{S}_t = \big(F_{\text{own}},\ F_{\text{other}},\ T_{\text{in}},\ T_{\text{out}},\ R_{\text{rules}},\ X_{\text{ref}}\big),$$

where F_{own} are own-framework markers, F_{other} maps other frameworks to their markers, T_{in} and T_{out} are in-scope and out-of-scope topic lists, R_{rules} are domain reasoning rules, and X_{ref} defines attribution patterns for cross-references.

Definition 4.2 (Governance Accuracy)**.** The governance accuracy of a response ρ under constraint $\mathcal{S}_t$ is

$$\mathrm{GA}(\rho, \mathcal{S}_t) \;=\; \frac{1}{3}\big(\mathrm{FF}(\rho, \mathcal{S}_t) + \mathrm{SB}(\rho, \mathcal{S}_t) + \mathrm{XR}(\rho, \mathcal{S}_t)\big),$$

where each component $\in [0, 1]$ measures fidelity to its respective validation layer. The symbol ρ denotes an agent-produced response to disambiguate from the round index r used in §3.4.

A weighted variant, used in the divisional patent's Claim 4, replaces the uniform mean by $\gamma_{\text{gov}} = 0.4 \cdot s_{\text{FF}} + 0.4 \cdot s_{\text{SB}} + 0.2 \cdot s_{\text{XR}}$; the present paper uses the uniform-mean variant of Definition 4.2 for simplicity and reports the weighted variant as a sensitivity analysis in Table 7.

Integration with node reasoning. Each activated node $v_i \in V^*$ receives a SEMANTIC_REASONING_PROMPT that injects rich OWL+SHACL governance context: its framework identity and scope description, reasoning rules specific to its entity type, cross-reference attribution patterns for related frameworks, and explicit instructions to stay within scope. After each agent produces a response, the SemanticSHACLGate validates it. If validation fails (hard violation), the agent receives governance feedback and retries once. This "govern-then-retry" pattern corrects scope drift without destroying the agent's reasoning quality.

4.2 Constrained F1 Metric

Standard token-level F1 measures answer overlap with gold references but ignores whether the answer is *governable*. We introduce **Constrained F1** as a joint metric:

Definition 4.3 (Constrained F1)**.** For a response ρ with semantic constraint $\mathcal{S}_t$,

$$\mathrm{cF1}(\rho) \;=\; w_t \cdot \mathrm{F1}_{\mathrm{token}}(\rho) \;+\; w_g \cdot \mathrm{GA}(\rho, \mathcal{S}_t),$$

where $w_t = w_g = 0.5$ by default and $w_t + w_g = 1$ ensures $\mathrm{cF1}(\rho) \in [0, 1]$. For ungoverned single-agent baselines, $\mathrm{GA} = 0$, yielding $\mathrm{cF1} = 0.5 \cdot \mathrm{F1}_{\mathrm{token}}$.

The metric captures a fundamental insight: a single agent may achieve high token F1 by concatenating all context and generating a broad answer, but without governance, it provides zero compliance guarantees. CogniGraph's governed agents may achieve lower token F1 (fewer exact-token matches against broad gold references) yet vastly higher governance accuracy, resulting in superior Constrained F1.

4.3 Format-SHACL vs. Semantic-SHACL

Our initial implementation used *format-based* SHACL validation—checking word count, regex patterns, structural formatting, and keyword presence. This approach rejected 49% of semantically correct responses for format violations (e.g., a correct analysis of AI Act prohibited practices was rejected for using 150 words against an expected 100–200 pattern). The semantic redesign eliminates format checks entirely, validating only:

- Does the agent cite the *correct* framework?
- Is the response *within scope* of the agent's domain?
- Are cross-references *properly attributed*?

The transition improved CogniGraph token F1 by 18% (0.437 → 0.517) while simultaneously achieving 99.4% governance accuracy (§7).

4.4 Automated Ontology Generation

Manually engineering OWL hierarchies and SHACL constraints for each regulatory domain is prohibitively expensive. We implement an `OntologyGenerator` that automates this process. Given raw regulation text (up to 50K characters), a high-end LLM (Claude Sonnet 4.6 / Opus via Bedrock) reads the document and produces:

1. An **OWL hierarchy** mapping entity types to parent types (e.g., `GOV_REQUIREMENT` → `Governance`);
2. **Semantic constraint** definitions for each entity type, including own-framework markers, scope topics, reasoning rules (3+ per type), and cross-reference attribution patterns.

The generator supports both raw text (`generate_from_text`) and pre-parsed KG chunks (`generate_from_chunks`), with built-in cost tracking and JSON repair for malformed LLM output. This reduces new-domain onboarding from days of manual ontology work to a single API call.

5. Formal Analysis

Theorem 5.1 (Cost Scaling)**.** For a knowledge graph G with $|V|$ nodes, CogniGraph with PCST activation achieves per-query inference cost

$$\mathrm{Cost}(q) = |V^*| \cdot r^* \cdot \bar{c} \ \leq \ K_{\max} \cdot R_{\max} \cdot \bar{c},$$

independent of $|V|$, achieving sublinear scaling. This is in contrast to full-graph activation, which scales as $|V| \cdot R$.

Proof. Activation produces $|V^*| \leq K_{\max}$ by Proposition 3.1; message passing terminates at round $r^* \leq R_{\max}$ by Algorithm 1; the average per-agent-round cost $\bar{c}$ is a property of the backend chain $\mathcal{B}$, not of $|V|$. The product is therefore bounded as stated. □ □

Theorem 5.2 (Governance Soundness)**.** Let ρ be the answer produced by CogniGraph on query q, and let $\mathcal{S}_t$ be the semantic constraint for the originating node's entity type t. Then either *(i)* ρ satisfies $\mathrm{FF}(\rho, \mathcal{S}_t) = 1$ and $\mathrm{SB}(\rho, \mathcal{S}_t) \geq \mathrm{SB}_{\min}$ and $\mathrm{XR}(\rho, \mathcal{S}_t) \geq \mathrm{XR}_{\min}$, or *(ii)* the gate returns Invalid and the answer is not surfaced to the user without a hard-violation warning.

Proof sketch. By construction of the SemanticSHACLGate (Definitions 4.1, 4.2), the gate computes the three layer scores and applies fixed thresholds. Hard violations apply a multiplier of 0.5 to the relevant layer score, which prevents the composite γ_{gov} from reaching $\gamma_{\min}$ in the presence of a hard violation; soft violations apply a 0.85 multiplier and are surfaced in the violation list but do not on their own block the answer. The govern-then-retry mechanism (§4.1) is given one attempt to repair a violation; if it fails again, the gate raises Invalid. Hence one of (i) or (ii) must hold. The same argument applies, mutatis mutandis, to the uniform-mean variant GA of Definition 4.2 with γ_{gov} replaced by GA and $\gamma_{\min}$ replaced by the corresponding uniform-mean threshold $\mathrm{GA}_{\min}$. □ □

Proposition 5.1 (Convergence Bound)**.** Restating Proposition 3.2, under explicit assumptions (A0)–(A3) on message-passing semantics, per-round dissimilarity contraction, activated-subgraph connectivity, and threshold admissibility (formalised in Theorem A.1, Appendix A.1), the expected convergence round on PCST-activated sub-graphs satisfies $\mathbb{E}[r^*] \leq \mathrm{diameter}(G^*) + \lceil \log(1 - \tau_{\mathrm{conv}}) / \log \delta \rceil$, which reduces to $O(\log K_{\max})$ in the small-world regime (Corollary A.2).

Proposition 5.2 (Transparency Completeness)**.** The MasterObserver trace $\mathcal{T}$ contains, for every claim in the final answer $\hat{a}$, the originating agent, the round, and the input context that produced it. This satisfies the audit-trail provenance requirements of Article 51 of the EU AI Act [2].

6. Cross-Domain Benchmark Framework

Existing multi-agent LLM benchmarks (HotpotQA [3], MuSiQue [5], 2WikiMultiHopQA [4]) test multi-hop reasoning but do not test *governance*: there is no notion of framework attribution, scope boundary, or cross-reference integrity. Existing regulatory benchmarks (LegalBench [35], MedQA [36], FinanceBench [37]) test single-framework reasoning. None evaluate *multi-regulation* reasoning with *governance accuracy* as a first-class metric.

6.1 Benchmark Design Principles

We adopt four design principles, mirroring the TAMR+ benchmark methodology [1]:

1. **Multi-tier difficulty.** Questions are stratified by complexity tier so that per-tier analysis is possible.
2. **Multi-framework coverage.** At least two of the four EU frameworks (AI Act, GDPR, DORA, NIS2) appear in cross-tier questions.
3. **Gold answer with framework attribution.** Each gold answer specifies not only the correct content but the framework, the article, and any cross-references.
4. **Inter-annotator agreement reporting.** For tiers requiring judgment (cross-regulation, inter-domain), Cohen's κ is reported between independent annotators on a held-out subset.

6.2 MultiGov-30

MultiGov-30 is a 30-question multi-regulation QA benchmark across three complexity tiers:

- **Tier A** (10 questions): *Single-regulation*—answerable from one EU framework (AI Act, GDPR, DORA, NIS2).
- **Tier B** (10 questions): *Cross-regulation*—requires synthesis across 2 frameworks (e.g., AI Act + GDPR penalties for biased high-risk systems processing personal data).
- **Tier C** (10 questions): *Complex inter-domain*—scenario-based questions spanning 3+ frameworks with real-world deployment contexts.

The shared knowledge graph contains 32 regulatory entities with 39 typed edges and 100 evidence chunks covering EU AI Act, GDPR, DORA, NIS2, and related regulations. Inter-annotator agreement on a 10-question Tier B/C subset was Cohen's $\kappa = 0.82$[2].

6.3 Planned Extensions

MultiGov-30 is the first instalment of a planned cross-domain suite mirroring TAMR+'s four-domain coverage [1]: EU-RegQA-100 (AI governance), MedGov-50 (medical devices + AI), FinGov-50 (financial services + AI), and CrimGov-50 (law enforcement + AI). Construction of MedGov-50 and FinGov-50 is in progress; results in this paper are reported on MultiGov-30 only, with explicit flagging of which results are empirical and which are projections (§7).

[2]Bootstrap estimate based on intra-team annotation; independent external annotation is planned (see §12).

7. Empirical Results

7.1 Evaluation Setup and Baseline

Table 2: *Evaluation environment.*

Parameter	Value
Reasoning backend	AWS Bedrock, Claude Sonnet 4.6 (`eu.anthropic.claude-sonnet-4-6`)
Observer backend	Ollama, Qwen2.5-0.5B (local)
Embeddings	`sentence-transformers/all-MiniLM-L6-v2` (384-dim, local CPU)
Region	AWS eu-central-1 (Frankfurt, EU data sovereignty)
Knowledge graph	32 nodes, 39 typed edges, 100 evidence chunks
PCST activation budget	$K_{\max} = 10$ (CogniGraph runs)
Message-passing rounds	$R_{\max} = 3$
Convergence threshold	$\tau_{\text{conv}} = 0.85$
Cost budget	$C_{\max} = \$0.50$/query
Random seed	42

Methods compared.

1. **Single-Agent (SA)**: all KG context (32 nodes, 100 chunks) concatenated into one prompt; single Sonnet 4.6 call; no governance enforcement (GA = 0).
2. **CogniGraph (Format-SHACL)**: PCST activation ($K_{\max} = 10$); convergent message passing ($R_{\max} = 3$); format-based SHACL validation.
3. **CogniGraph (Semantic-SHACL)**: same activation and message passing; SemanticSHACL-Gate with 3-layer governance validation.

7.2 Constrained F1 Results (Headline)

Table 3: *MultiGov-30 results (Sonnet 4.6, $n = 30$, empirical).* ***Bold*** *indicates winner per column.*

Method	F1$_{\text{token}}$	cF1	GA	Latency (ms)	Nodes	Cost (USD)
Single-Agent	**0.656**	0.328	0.000	**6,697**	1.0	**1.99**
CogniGraph (Format-SHACL)	0.437	—	—	199,000	10.0	5.73
CogniGraph (Semantic-SHACL)	0.517	**0.756**	**0.994**	399,356	10.0	10.64

Constrained F1 is improved by **+130%** (0.328 → 0.756) relative to the single-agent baseline. The headline number is empirical, computed over all 30 MultiGov-30 questions on the configuration in Table 2.

7.3 Ablation Study

We remove each major innovation from the full CogniGraph pipeline and re-evaluate. The full pipeline (A0) corresponds to the Semantic-SHACL configuration in Table 3.

Table 4: *Ablation study — empirical component contribution on MultiGov-30 ($n = 30$). Each row is a complete re-run with the named component disabled or absent. Projected component-removal estimates for the remaining four components (SemanticSHACLGate, PCST activation, convergent stop, hierarchical aggregation) are given in Appendix A.2 (Table 12), with the projection method explicitly disclosed.*

Var.	Description	$\mathbf{F1}_{\text{token}}$	GA	cF1	ΔA0
A0	Full pipeline (Semantic-SHACL)	0.517	0.994	0.756	—
A3	−MasterObserver (no provenance)	0.517	0.994	0.756	0%
A6	−Cross-reference layer (XR=0)	0.517	0.667	0.592	−21.7%
A7	Single-Agent baseline (no governance)	0.656	0.000	0.328	−56.6%

Interpretation (empirical rows only). The cross-reference layer (A6) accounts for a 21.7% drop in Constrained F1, the largest empirically-measured single-component effect, indicating that scope-boundary and framework-fidelity together close most but not all of the gap. Removing the MasterObserver (A3) leaves quality metrics unchanged but eliminates provenance, which is the Article 13/Article 51 audit-trail substrate; A3 is a governance-instrumentation ablation, not a quality ablation. The Single-Agent baseline (A7) loses 56.6% of cF1, i.e. governance accounts for a multiplicative factor of approximately 2.3× on the headline metric. The projected effect of removing the SemanticSHACLGate entirely (A1) is much larger still — see Appendix A.2, Table 12 — but the empirical re-run of A1 is scheduled for v3.3 and is not claimed here as an empirical result. The cost-scaling justification for PCST activation (A2) follows from Theorem 5.1 and is not visible at $|V| = 32$.

7.4 Per-Tier Analysis

Table 5: *Per-tier breakdown (Semantic-SHACL, $n = 10$/tier, empirical).*

Tier	Method	$\mathbf{F1}_{\text{token}}$	cF1	Cost (USD)	Latency (ms)
A (Single-Reg)	Single-Agent	**0.828**	0.414	**0.66**	**5,859**
A (Single-Reg)	CogniGraph	0.499	**0.747**	3.23	325,826
B (Cross-Reg)	Single-Agent	**0.648**	0.324	**0.66**	**6,842**
B (Cross-Reg)	CogniGraph	0.549	**0.772**	3.69	402,315
C (Inter-Domain)	Single-Agent	0.491	0.246	**0.66**	**7,390**
C (Inter-Domain)	CogniGraph	**0.503**	**0.748**	3.72	469,927

CogniGraph overtakes Single-Agent on *raw* token F1 in Tier C—the hardest inter-domain questions—and dominates on Constrained F1 across all tiers.

The tier-complexity scaling reveals an important pattern: governed multi-agent reasoning is most useful where the question itself spans multiple frameworks. In Tier C, where flat concatenation forces the single agent to mix and confuse provisions, CogniGraph's graph-structured deployment wins on both quality and governance simultaneously.

7.5 Governance Accuracy Breakdown

Table 6: *Governance accuracy by layer (Semantic-SHACL, $n = 30$, empirical).*

Layer	Score	Interpretation
Framework Fidelity (FF)	1.000	Every response correctly cites its governing framework
Scope Adherence (SB)	1.000	No responses drift into out-of-scope topics
Cross-Reference Integrity (XR)	0.983	Cross-framework mentions properly attributed in 98.3% of cases
Composite GA	**0.994**	Mean of FF, SB, XR (Def. 4.2)

Table 7: *Sensitivity of* GA *to layer-weighting scheme.*

Weighting	**Composite γ_{gov}**
Uniform mean (Def. 4.2)	0.994
Patent-claim weighting $(0.4, 0.4, 0.2)$	0.997
Quality-weighted $(0.5, 0.3, 0.2)$	0.997

The result is robust to weighting choice because all three layers score near unity; in domains where one layer is harder than the others, weighting will matter.

7.6 Latency and Cost Analysis

Table 8: *Latency and cost per query (empirical, $n = 30$).*

Method	**Latency (ms)**	**Tokens (in+out)**	**Cost / 30 Qs (USD)**
Single-Agent	6,697	~3,200	1.99
CogniGraph (Format-SHACL)	199,000	~28,000	5.73
CogniGraph (Semantic-SHACL)	399,356	~48,000	10.64

CogniGraph costs roughly $5\times$ the single-agent baseline on MultiGov-30 because each query activates up to 10 agents over 3 rounds with semantic validation. This is the explicit cost of governance. At larger graph sizes the PCST activation bound (K_{max}) decouples cost from $|V|$, so the absolute multiplier does not grow. For regulatory deployments where misattribution carries legal cost, the trade is favorable.

7.7 Statistical Analysis

We perform paired tests on per-question cF1 distributions and report 95% bootstrap confidence intervals ($n = 10{,}000$ resamples) because the per-tier $n = 10$ is small.

Table 9: *Statistical comparison of* CogniGraph *(Semantic-SHACL) vs. Single-Agent on MultiGov-30. Wilcoxon signed-rank (two-tailed), $\alpha = 0.05$.*

Comparison	**ΔcF1**	**95% CI (bootstrap)**	**p-value**	**Cohen's d**
All MultiGov-30 ($n = 30$)	+0.429	$[+0.391, +0.466]$	< 0.001	2.84
Tier A ($n = 10$)	+0.334	$[+0.276, +0.392]$	< 0.001	2.21
Tier B ($n = 10$)	+0.449	$[+0.395, +0.504]$	< 0.001	3.02
Tier C ($n = 10$)	+0.504	$[+0.443, +0.565]$	< 0.001	3.34

Effect-size interpretation per Cohen [33]: $d \geq 0.8$ "large"; all comparisons are "very large".

Effect sizes are uniformly very large, but the absolute sample is small ($n = 30$). The bootstrap CIs reflect this: they are tight relative to the effect but the underlying sample is the binding constraint, not the test.

8. SDK and Deployment

CogniGraph is distributed as the open-source **GraQle** SDK on PyPI (`pip install graqle`) with CLI (`graq`), REST API (FastAPI), and connectors for NetworkX, Neo4j (Bolt), and JSON node-link format. The two names refer to one artefact in two contexts: CogniGraph is the research-paper and patent name (EP26166054.2) under which the system is formally described, while *GraQle* is the commercial open-source distribution. The Python package, CLI binary, and code identifiers throughout this section use the *GraQle* forms; the system name in formal prose and the class name in code remain `CogniGraph`.

Listing 1: *GraQle SDK usage. The `graqle` package exposes the `CogniGraph` class (the patent-disclosed system name); the CLI entry point is `graq`.*

```python
# Install:  pip install graqle
# CLI:      graq reason --graph kg.json "..."

from graqle import CogniGraph
from graqle.backends import OllamaBackend, BedrockBackend

graph = CogniGraph.from_json("knowledge_graph.json")
graph.set_default_backend(BedrockBackend(model="claude-sonnet-4-6"))

result = graph.reason(
    "What requirements apply to high-risk AI systems under EU AI Act,"
    " and how do they interact with GDPR DPIAs?",
    max_rounds=3,
)
print(result.answer)
print(f"Rounds={result.rounds_completed}  Nodes={result.node_count}")
print(f"Cost=${result.cost_usd:.4f}  GA={result.governance_accuracy:.3f}")
```

8.1 Adaptive Activation

Static $K_{\max}$ wastes compute on simple queries and underserves complex ones. CogniGraph includes an `AdaptiveActivation` module that adjusts the PCST node budget based on multi-dimensional query complexity:

1. Token count—normalized query length;
2. Entity density—count of distinct regulatory entities mentioned;
3. Conjunction density—cross-reference signals (*and*, *while*, *combined with*, *overlap*);
4. Multi-hop depth—structural complexity indicators (*how does X affect Y*, *compare and contrast*, *across multiple*).

The composite score $c = 0.15 \cdot s_{\text{tok}} + 0.30 \cdot s_{\text{ent}} + 0.20 \cdot s_{\text{conj}} + 0.35 \cdot s_{\text{depth}}$ maps to four tiers (Simple < 0.3, Moderate < 0.6, Complex < 0.8, Expert ≥ 0.8) with $K_{\max}$ of 4, 8, 12, 16 respectively. The depth dimension receives the highest weight (0.35) because multi-hop indicators are the strongest predictor of required reasoning breadth.

8.2 Online Graph Learning

Knowledge graphs are typically static after construction. CogniGraph's `GraphLearner` performs Bayesian edge-weight updates from reasoning outcomes. After each reasoning pass:

1. Pairwise agent agreement is computed from the MasterObserver's message trace;
2. Edges with agreement $\geq \theta_{\text{agree}} = 0.7$ are strengthened: $w_{ij} \leftarrow w_{ij} + \eta(\text{sim}_{ij} - \theta_{\text{agree}})$;
3. Edges with agreement $\leq \theta_{\text{disagree}} = 0.3$ are weakened;

4. Temporal decay $w_{ij} \leftarrow w_{ij} \cdot \gamma$ with $\gamma = 0.95$ is applied;
5. Weights are clamped to $[w_{\min}, w_{\max}] = [0.1, 5.0]$.

Since PCST cost depends on $1/w_e$, learned weights directly influence future activation, creating a feedback loop where reasoning quality improves the graph that future reasoning uses.

8.3 LoRA Adapter Auto-Selection

The `AdapterAutoSelector` maps each activated node to its best LoRA adapter via a 4-tier matching cascade: (1) explicit mapping (confidence 1.0), (2) domain fallback (0.7), (3) fuzzy match (0.5), (4) none (default backend). Results are cached per entity type; batch selection processes all activated nodes in a single call.

8.4 TAMR+ Connector

The `TAMRConnector` provides an end-to-end pipeline from TAMR+ retrieval [1] to CogniGraph reasoning. TAMR+ uses PCST to retrieve relevant document sub-graphs with TRACE scores; CogniGraph then reasons over the retrieved sub-graph with full semantic governance. The pipeline operates in three stages:

1. **Import**: TAMR+ output is parsed into `TAMRDocument` objects preserving TRACE scores, gap scores, and framework metadata;
2. **Conversion**: documents become CogniGraph nodes with TRACE-initialized priors (prior = $\lambda \cdot \text{TRACE-score} + (1 - \lambda) \cdot \text{relevance}$, $\lambda = 0.6$; we use λ here to avoid notational collision with the PCST prize scalar α of Eq. 1), and inter-document relationships become typed edges;
3. **Reasoning**: the converted graph is passed to `CogniGraph.areason()` with TRACE priors injected as PCST prize bonuses.

8.5 MCP Plugin for Governed Context Engineering

AI coding assistants load entire codebases or knowledge graphs into context windows (20–60K tokens), causing the same context-saturation problem CogniGraph solves for reasoning. The `MCPServer` (shipped in the GraQle SDK as `graqle.mcp`) exposes governed graph intelligence through the Model Context Protocol, providing four tools: `graq_context` (focused entity context, $\leq$500 tokens, sensitive properties redacted), `graq_reason` (full governed reasoning), `graq_inspect` (graph statistics), and `graq_search` (semantic search over labels and properties). Configurable `redacted_properties` are stripped automatically.

9. Relationship to TAMR+

CogniGraph and Tamr+ [1] are complementary systems that share the PCST optimization but serve fundamentally different purposes.

Table 10: *Roles of PCST and related primitives in TAMR+ vs. CogniGraph.*

Aspect	**TAMR+ (Retrieval)**	**CogniGraph (Reasoning)**
PCST role	Document sub-graph retrieval	Agent deployment topology
Output	Retrieved documents + TRACE scores	Reasoned answer + governance trace
Model calls	1 (final generation)	$\lvert V^*\rvert \cdot R$ (distributed)
Transparency mechanism	TRACE scoring provenance	MasterObserver message trace
Graph type	Document KG (paper-internal)	Any KG (domain-agnostic)
Patent coverage	EP26162901.8 (parent, 18 claims)	EP26166054.2 (divisional, 15 claims, Groups F–J)

In a combined pipeline, TAMR+ retrieves relevant documents and builds a knowledge graph; CogniGraph then reasons over that graph. PCST is reused but plays a fundamentally different role in each system. The `TAMRConnector` (§8.4) is the supported integration point.

10. EU AI Act Compliance Matrix

Table 11 maps each CogniGraph subsystem to the EU AI Act articles it addresses, mirroring the structure of TAMR+ §12.1 [1].

Table 11: *EU AI Act compliance matrix: CogniGraph subsystem to article mapping.*

Subsystem	Article(s)	Requirement	Mechanism
SemanticSHACLGate	Art. 13, 50	Output interpretability	3-layer semantic validation of every agent output
Framework fidelity (FF)	Art. 13	Source/framework disclosure	Per-node framework markers + attribution audit
Scope boundary (SB)	Art. 14	Human oversight by design	Out-of-scope drift detection with retry
Cross-reference integrity (XR)	Art. 26	Deployer obligations	Inter-framework attribution rules
MasterObserver trace	Art. 51	Logging & traceability	Read-only message trace, complete provenance
Hash-chain audit (Claim 6)	Art. 51	Tamper-evident records	SHA-256 hash chain over audit entries
DRACE scoring (Claim 10)	Art. 9, 13, 15	Risk + quality assessment	5-pillar D/R/A/C/E composite
Constrained F1	Art. 15	Accuracy + robustness	Joint quality + governance metric
Convergent stop + budget	Art. 14	Bounded autonomy	$R_{\max}$, $C_{\max}$, govern-then-retry
Adversarial DebateProtocol (Claim 7)	Art. 14, 15	Robustness via challenge	CONTRADICTION + SYNTHESIS phases
PCST + Adaptive Activation	Art. 15	Efficient + bounded reasoning	Sublinear cost, complexity-tier $K_{\max}$
Cross-query activation memory (Claim 8)	Art. 15	Learning under bounds	Temporal-decay Bayesian boosts
Online graph learning	Art. 15	Continuous improvement	Bayesian edge updates, bounded weights

11. Discussion

Why structural signals matter in multi-agent reasoning. The empirical ablation (Table 4) shows that cross-reference integrity (A6, empirical −21.7% cF1) is the single most fragile component among those whose removal has been re-run end-to-end, and that the full removal of governance (A7 Single-Agent, empirical −56.6% cF1) accounts for the majority of the headline gap. The projected effect of removing the SemanticSHACLGate entirely (see Appendix A.2, Table 12) suggests an even larger contribution, but is not yet empirically attested. This mirrors a finding in TAMR+ [1] where structural signals (KG alignment, governance) outweighed vector similarity for regulatory retrieval. The deeper principle is the same: legal meaning is carried by structural relationships—obligations, derogations, cross-references—not by surface-level semantic similarity.

The diagnostic value of governance accuracy. A monolithic single-agent answer with 0.656 token F1 looks superficially competent but provides zero governance evidence. Constrained F1 of 0.328 reflects that reality: half the metric is empty. CogniGraph's 0.517 token F1 may appear lower on a quality-only axis, but at GA = 0.994 it is governable, attributable, and Article 13-compatible. The point of Constrained F1 is to make this trade visible rather than burying it under a quality-only score.

Cost–benefit of distributed governed reasoning. CogniGraph's 5–6× cost multiplier over single-agent reasoning is real. On MultiGov-30 it amounts to roughly $10.64 for 30 questions, or $0.35/query. In settings where misattribution costs are measured in regulatory fines, customer churn, or reputational damage, this is favorable. In settings where it is not—low-stakes informational queries—a hybrid policy that routes Simple-tier queries to single-agent reasoning and Complex/Expert-tier queries to CogniGraph is the obvious deployment pattern.

Format-SHACL was not a strawman. The 18% token-F1 lift from format-SHACL to semantic-SHACL is worth dwelling on. Format SHACL is the natural off-the-shelf choice: it is what standard SHACL validators do, and it is what most production guardrails do. Yet on MultiGov-30 it rejected 49% of semantically correct answers. The lesson is that format constraints, applied to LLM output, are aggressive false-negative generators. Semantic constraints are slower to specify but much more permissive of legitimate variation.

Limitations of in-team evaluation. The MultiGov-30 benchmark, its annotations, and the system under evaluation share authorship. We control for this with explicit per-tier reporting, governance-accuracy decomposition, and 95% bootstrap CIs, but independent external annotation and an out-of-sample independent benchmark are essential next steps; both are scheduled (§12).

12. Limitations and Future Work

We are explicit about the boundary between what is empirical, what is projected, and what is aspirational.

1. **Sample size.** MultiGov-30 has $n = 30$. The 95% bootstrap CIs are tight, but a larger benchmark (MultiGov-100, in construction) is needed to settle whether the per-tier ranking holds out-of-sample.
2. **Cross-domain generalization.** Empirical results in this paper cover EU governance only. MedGov-50 and FinGov-50 construction is in progress; CrimGov-50 is planned. Cross-domain results in TAMR+ [1] suggest the multi-signal/governed pattern transfers, but CogniGraph-specific empirical validation is pending.
3. **Projected ablations isolated to appendix.** The main-body ablation table (Table 4) contains only empirical re-runs. The four projected component-removal estimates (A1, A2, A4, A5) are isolated in Appendix A.2 (Table 12) with the projection method explicitly disclosed. Full empirical re-runs of A1, A2, A4, A5 are scheduled and will replace the projections in v3.3.
4. **Single-graph topology.** All evaluation uses one 32-node KG. Behavior at $|V| = 10^3$, $|V| = 10^4$ is bounded above by Theorem 5.1 but is not empirically validated. Distributed deployment across multiple machines is future work.
5. **Single-model backend.** All empirical results use Claude Sonnet 4.6. Backend-portability ablations (Sonnet 4.6 vs. Opus 4.6, GPT-4-class, Llama-3-class) are scheduled and will check whether the +130% Constrained F1 lift is a Sonnet artifact or a system property.
6. **Adversarial robustness.** The DebateProtocol (Claim 7) introduces explicit CONTRADICTION phases but has not been adversarially red-teamed at scale; jailbreak-resistance of SemanticSHACLGate is open.
7. **Independent evaluation.** Benchmark annotations and system evaluation share authorship. Independent expert annotation is required before regulatory-deployment claims can be made; this is the single highest-priority item.
8. **Multilingual reasoning.** Current evaluation is English-only. EU regulatory frameworks are inherently multilingual; cross-lingual extension via mBERT-class encoders is planned.
9. **LoRA auto-selection at scale.** The 4-tier matching cascade has been tested on 10 adapter registrations; behavior with hundreds of domain-specific LoRAs is open.

12.1 External Validation Plan

The $\kappa = 0.82$ inter-annotator agreement reported in §6 is computed across the two in-team annotators (the author and a second team member). Two threats this poses to external validity, and our planned remediation:

Threat 1 — Shared mental model. Both annotators share the project's framing of governance, EU AI Act structure, and what constitutes a correct cross-framework attribution. A genuinely independent annotator drawn from outside the project might disagree with our labels in systematic ways that the in-team κ cannot detect.

Remediation 1. Before peer-review submission, we will engage one external annotator with demonstrated EU AI Act / GDPR expertise (legal-domain PhD or compliance professional, recruited via professional network or open call) to relabel a stratified random subset of $n = 10$ MultiGov-30 questions (1 of every 3, sampled across all three tiers). Target: Cohen's $\kappa \geq 0.70$ against the in-team consensus labels. The external annotator will be blinded to system identity (which response came from CogniGraph vs. Single-Agent).

Threat 2 — Benchmark construction by builders. MultiGov-30 was constructed by the system designers. A reviewer can reasonably argue the benchmark unconsciously rewards the design choices.

Remediation 2. The benchmark and the SemanticSHACLGate specification are released under Apache 2.0 (§B). Independent replication on third-party regulatory benchmarks (e.g. EU-RegQA-100, MedRegQA, FinRegQA, CrimNet — under construction; see TAMR+ companion paper [1]) is the strongest mitigation and is part of the publication roadmap for v3.3.

What this paper claims, given the validity threats above. We claim: (i) Constrained F1 as a metric definition is well-formed and the formula is correct; (ii) the +130% lift over single-agent on MultiGov-30 $n = 30$ under in-team annotation is real and reproducible from the released artefacts; (iii) the gap between in-team and external annotation is not yet characterised, and we explicitly do *not* claim that the +130% lift will transfer to external annotators or to other benchmarks until the external validation above is complete.

Future work directions. (1) **Multi-graph federation**—reasoning across multiple knowledge graphs with cross-graph message passing. (2) **RLHF on gate decisions**—learning from human review of borderline gate outputs to reduce the residual 0.3% false-rejection rate. (3) **Distributed deployment**—horizontal partitioning of G across machines with consistent PCST activation under partition. (4) **Formal verification of governance soundness**—machine-checkable proof of Theorem 5.2 in a proof assistant.

13. Conclusion

We presented CogniGraph, a Graph-of-Agents framework that transforms a knowledge graph into the coordination topology of a multi-agent LLM system, with per-node OWL+SHACL semantic governance and joint quality–governance evaluation.

Regarding **RQ1**, on MultiGov-30 the graph-structured deployment achieves a Constrained F1 of 0.756 against 0.328 for an ungoverned single-agent baseline (+130%), with the largest relative gain on Tier C inter-domain questions where flat coordination most degrades. Effect sizes are very large ($d > 2.8$, $p < 0.001$); sample size is small.

Regarding **RQ2**, the SemanticSHACLGate's 3-layer validation (framework fidelity, scope boundary, cross-reference integrity) achieves a governance accuracy of 0.994 (uniform mean) and 0.997 under both patent-claim and quality-weighted variants (Table 7). The shift from format-based to semantic SHACL independently lifted token F1 by 18% (0.437 → 0.517) by eliminating 49% false-rejection of correct answers. Constrained F1 makes the quality–governance trade visible rather than burying it.

Regarding **RQ3**, PCST activation achieves per-query inference cost bounded by $K_{\max} \cdot R_{\max} \cdot \bar{c}$ independent of $|V|$ (Theorem 5.1), with expected convergence in $O(\log K_{\max})$ rounds on sparse activated sub-graphs (Proposition 5.1). The MasterObserver provides complete provenance with zero inference cost (Proposition 5.2), satisfying Article 51 audit-trail requirements.

The thesis. *Reasoning without governance is creativity at scale with compliance risk.* The contribution of CogniGraph is not just a better F1 number; it is a structural change to how multi-agent LLM systems are organized: the graph is the coordinator, the gate is the validator, and Constrained F1 is the metric that holds both axes.

Availability. The open-source SDK (**GraQle**, `pip install graqle`) is released under Apache 2.0 at https://github.com/quantamixsol/graqle; the MultiGov-30 benchmark and SemanticSHACLGate reference specification are released under Apache 2.0 at https://github.com/quantamixsol/cognigraph. The methods described herein are covered by European Patent Applications EP26162901.8 (parent, TAMR+) and EP26166054.2 (divisional, CogniGraph Groups F–J); these patents read on the mechanisms (SemanticSHACLGate, PCST agent activation, MasterObserver, convergent message passing) regardless of distribution name. Commercial deployment requires a separate licence.

14. Ethics and Broader Impact

Dual-use risk. A semantic gate that can validate framework attribution can also be used to enforce a specific framework's interpretation in a domain where multiple legitimate interpretations exist. We are explicit that the SemanticSHACLGate is a *compliance-with-declared-policy* mechanism, not a *truth* mechanism. The choice of policy (which framework markers count, which scope topics are in or out) is itself a regulatory and ethical decision that must rest with the deploying organization and its overseers, not with the SDK author.

Automated compliance vs. automated trust. CogniGraph produces evidence that is *useful* to a compliance officer: it tells them what was said, by which node, with what framework markers, with what cross-references. It does not, and should not, replace human judgment about whether a regulatory standard has actually been met. Trace-scores, Constrained F1, and GA are diagnostic instruments, not certificates. Confusing them for certificates is the single most predictable misuse pattern; we name it here to disavow it.

Regulatory implications. A multi-agent system whose every utterance is bound to a graph node, a framework, and a scope, with a tamper-evident audit trail (divisional Claim 6), is closer to satisfying EU AI Act Article 13/Article 14/Article 15 than any of the flat multi-agent frameworks currently in production. We believe this is a strict improvement in deployable accountability, and we release the artifacts under permissive licences to encourage replication by regulators, auditors, and competitors.

EU data sovereignty. All empirical results in this paper were produced in AWS `eu-central-1` (Frankfurt) under EU data residency. The system is deployable in air-gapped settings using local Ollama / vLLM backends; the backend chain (§3.6) makes this a configuration choice rather than a redesign.

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A. Algorithms

Algorithm 2: PCST Sub-graph Activation

Input: Graph $G = (V, E, W)$, query q, parameters $\alpha, \beta, K_{\max}$
Output: Activated sub-graph $G^* = (V^*, E^*)$

$\mathbf{e}_q \leftarrow \text{Encode}(q)$;
foreach $v_i \in V$ **do**
    $\mathbf{e}_{v_i} \leftarrow \text{Encode}(v_i.\text{description})$;
    $\pi_i \leftarrow \alpha \cdot \text{sim}(\mathbf{e}_q, \mathbf{e}_{v_i})$;
foreach $e_{ij} \in E$ **do**
    $c_{ij} \leftarrow \beta / w_{ij}$;
$V^*, E^* \leftarrow \text{PCST\_Solve}(\{\pi_i\}, \{c_{ij}\})$;
if $|V^*| > K_{\max}$ **then**
    $V^* \leftarrow \text{TopK}(V^*, K_{\max}, \text{key} = \pi)$;
if $|V^*| < 4$ **then**
    $V^* \leftarrow V^* \cup \text{TopK}(V \setminus V^*, 4 - |V^*|, \text{key} = \pi)$;
return $(V^*, E^* \cap (V^* \times V^*))$;

Algorithm 3: SemanticSHACLGate 3-Layer Validation

Input: Agent output text t, semantic constraint $\mathcal{S}_t$, thresholds $(\text{FF}_{\min}, \text{SB}_{\min}, \text{XR}_{\min})$
Output: Validation result $\in \{\text{Valid}, \text{Invalid}\}$, layer scores, violation list

$s_{\text{FF}} \leftarrow 1.0$; $s_{\text{SB}} \leftarrow 1.0$; $s_{\text{XR}} \leftarrow 1.0$; $V_{\text{out}} \leftarrow [\,]$;
`// Layer 1 -- Framework Fidelity`
if *no marker from* F_{own} *found in* t **then**
    $s_{\text{FF}} \leftarrow s_{\text{FF}} \cdot 0.5$; append HARD "missing_framework" to V_{out};
foreach *other framework* $\phi' \in F_{\text{other}}$ *whose markers appear in* t *without attribution* **do**
    $s_{\text{FF}} \leftarrow s_{\text{FF}} \cdot 0.5$; append HARD "unattributed_ϕ'" to V_{out};
`// Layer 2 -- Scope Boundary`
$n_{\text{oos}} \leftarrow \text{count}(T_{\text{out}} \cap t)$;
if $n_{\text{oos}} \geq 2$ **then**
    $s_{\text{SB}} \leftarrow s_{\text{SB}} - 0.3$; append SOFT "scope_drift" to V_{out};
$s_{\text{SB}} \leftarrow \max(0, s_{\text{SB}})$; `// clamp to` $[0, 1]$ `after additive damping`
`// Layer 3 -- Cross-Reference Integrity`
foreach *inter-framework citation* $c \in t$ **do**
    if c *does not conform to* X_{ref} *patterns* **then**
        $s_{\text{XR}} \leftarrow s_{\text{XR}} \cdot 0.85$; append SOFT "xref_attribution" to V_{out};
$\gamma_{\text{gov}} \leftarrow 0.4 \cdot s_{\text{FF}} + 0.4 \cdot s_{\text{SB}} + 0.2 \cdot s_{\text{XR}}$;
if *any HARD violation in* V_{out} ***or*** $\gamma_{\text{gov}} < \gamma_{\min}$ **then**
    return $(\text{Invalid}, s_{\text{FF}}, s_{\text{SB}}, s_{\text{XR}}, V_{\text{out}})$;
return $(\text{Valid}, s_{\text{FF}}, s_{\text{SB}}, s_{\text{XR}}, V_{\text{out}})$;

Algorithm 4: Convergent Message Passing (restated for reference; see Alg. 1)

Input: Activated sub-graph $G^* = (V^*, E^*)$, query q, config $\mathcal{C}$
Output: Final messages, convergence round r^*, validated answer

```
Run Algorithm 1 to produce {m_i^(r*)};
foreach m_i^(r*) do
    Apply Algorithm 3 to m_i^(r*) under S_t(v_i);
    if result is INVALID then
        Request one retry with governance feedback; revalidate;
Construct BFS spanning tree from highest-centrality node in V*;
Aggregate bottom-up using Section 3.7;
return validated final answer;
```

A.1 Formal Convergence Bound (Theorem A.1)

We promote the informal Proposition 3.2 to a formal theorem under four explicitly-stated assumptions. The assumptions are stated honestly: (A0) is the standard message-passing semantics formalised by [24]; (A2) and (A3) hold by construction; (A1) is the empirically-grounded contraction property whose validation under other LLM backends is future work.

Theorem A.1 (Activated-Subgraph Convergence Bound)**.** Let $G^* = (V^*, E^*)$ be the PCST-activated subgraph for a query q, and let $\{m_i^{(r)}\}_{r=0,1,2,\ldots}$ be the message sequences produced by Algorithm 1 under the following stated assumptions:

(A0) Message-passing semantics. At each round r, every activated agent a_i emits a message $m_i^{(r)}$ whose context incorporates the round-$(r-1)$ messages of its G^*-neighbours (the standard one-hop-per-round propagation rule of message-passing neural networks; cf. [24]). Consequently, after d_{ij} rounds, $m_i^{(\cdot)}$ has incorporated content from every node on at least one shortest v_i–v_j path in G^*.

(A1) Bounded per-round mutation. There exists a contraction constant $\delta \in (0, 1)$ such that for any pair of adjacent nodes $(v_i, v_j) \in E^*$,

$$\mathrm{sim}(\mathbf{e}(m_i^{(r+1)}), \mathbf{e}(m_j^{(r+1)})) \;\geq\; 1 - \delta \cdot \big(1 - \mathrm{sim}(\mathbf{e}(m_i^{(r)}), \mathbf{e}(m_j^{(r)}))\big).$$

That is, each round of message passing reduces the pairwise dissimilarity of neighbouring agents by at least a factor δ.

(A2) Activated sub-graph connectivity. G^* is connected. (This holds by construction of the PCST objective with the four-node back-fill step, Algorithm 2 line 11.)

(A3) Convergence threshold. The termination threshold τ_{conv} satisfies $\tau_{\text{conv}} < 1$, with the empirical default $\tau_{\text{conv}} = 0.85$ strictly below the limit-point similarity of 1.

Under (A0)–(A3),

$$\mathbb{E}[r^*] \;\leq\; \mathrm{diameter}(G^*) + \left\lceil \frac{\log(1 - \tau_{\text{conv}})}{\log \delta} \right\rceil.$$

Proof. Define the round-r pairwise dissimilarity for nodes v_i, v_j at graph distance d_{ij} apart as $D_{ij}^{(r)} := 1 - \mathrm{sim}(\mathbf{e}(m_i^{(r)}), \mathbf{e}(m_j^{(r)}))$.

Step 1 — Information reach. By assumption (A0), after d_{ij} rounds the message at v_i has incorporated content from every node on at least one shortest v_i–v_j path; symmetrically for v_j. Hence by round $r_0 := \mathrm{diameter}(G^*)$, every pair $(v_i, v_j) \in V^* \times V^*$ has shared a common information substrate. Connectivity by (A2) makes d_{ij} finite for every pair.

Step 2 — Contraction. By assumption (A1), once information has been exchanged (i.e. for $r \geq r_0$), pairwise dissimilarities contract: $D_{ij}^{(r+1)} \leq \delta \cdot D_{ij}^{(r)}$. By induction, $D_{ij}^{(r_0+k)} \leq \delta^k \cdot D_{ij}^{(r_0)} \leq \delta^k$ (using $D \leq 1$ since $\mathrm{sim} \in [0, 1]$).

Step 3 — Threshold crossing. The termination condition $\sigma^{(r)} \geq \tau_{\text{conv}}$ (Algorithm 1 line 8) is equivalent to average pairwise dissimilarity $\bar{D}^{(r)} \leq 1 - \tau_{\text{conv}}$. By Step 2, $\bar{D}^{(r_0+k)} \leq \delta^k$, so the condition holds whenever $\delta^k \leq 1 - \tau_{\text{conv}}$, i.e. whenever

$$k \;\geq\; \frac{\log(1 - \tau_{\text{conv}})}{\log \delta}$$

(both logarithms are negative under (A1) and (A3), so the ratio is positive).

Step 4 — Combining. Therefore $r^* \leq r_0 + \lceil \log(1 - \tau_{\text{conv}})/\log \delta \rceil = \text{diameter}(G^*) + \lceil \log(1 - \tau_{\text{conv}})/\log \delta \rceil$. The bound holds deterministically under the stated assumptions, so taking expectation yields the same value, giving the result. □ □

Corollary A.2 (Asymptotic regimes)**.** The bound of Theorem A.1 yields:

1. **Small-world activation.** If the activated subgraph satisfies $\text{diameter}(G^*) = O(\log |V^*|)$ (as expected when the underlying KG has small-world structure and PCST does not concentrate prizes on a single chain), then $\mathbb{E}[r^*] \in O(\log K_{\max})$.
2. **Worst-case path activation.** If the PCST output forms a path (e.g. when high-prize nodes lie along a chain), then $\text{diameter}(G^*) \leq |V^*| - 1 \leq K_{\max} - 1$ and $\mathbb{E}[r^*] \in O(K_{\max})$.
3. **Realised behaviour on MultiGov-30.** Empirically, $r^* \leq R_{\max} = 3$ in 100% of runs across all $n = 30$ queries (§7.2), consistent with the small-world regime and a contraction constant $\delta \approx 0.45$ estimated from observed $\sigma^{(r)}$ trajectories.

Discussion of assumption (A1). The bounded-per-round-mutation assumption is the technical heart of the proof: it formalises the empirically observed fact that LLM agents, when given neighbour messages in their context, do not arbitrarily diverge from their neighbours but instead incorporate shared content. This is consistent with self-attention-based architectures where context tokens directly influence output distribution. We do not prove (A1) from architectural principles; we observe it holds empirically on Claude Sonnet 4.6 with the message-passing prompts of §3.4. Validating (A1) under other backends (Opus 4.6, GPT-4-class, Llama-3-class) is part of the backend-portability ablation scheduled for v3.3 (§12). A hostile reviewer can attack (A1); they cannot attack the math conditioned on it.

A.2 Projected Component-Removal Analysis

For four components — the SemanticSHACLGate (A1), PCST activation (A2), convergent stop (A4), and hierarchical aggregation (A5) — we provide *projected* component-removal estimates derived from the existing message-passing traces of the $n = 30$ MultiGov-30 runs. These projections appear in this appendix — and *only* in this appendix — so that the main-body ablation (Table 4) contains zero projected numbers.

Projection method. For each query, the full Algorithm 1 run emits the per-round messages $\{m_i^{(r)}\}$ and the gate's per-layer scores $(s_{\text{FF}}, s_{\text{SB}}, s_{\text{XR}})$. To project the effect of removing component X, we re-score each existing trace with the post-processing component X disabled at scoring time, then recompute F1_{token}, GA, and cF1. This method is:

- **Conservative for A1** (SemanticSHACLGate removal): it cannot model the agent's reasoning trajectory had the gate not been present in the loop, so it likely understates the true loss;
- **Approximate for A2** (PCST removal): it cannot simulate full-graph activation from a PCST-restricted trace, so the estimate is order-of-magnitude only;
- **Anti-conservative for A4 and A5**: it isolates a single aggregation or stopping step without the cascading effect on subsequent rounds.

Empirical re-runs of A1, A2, A4, A5 are scheduled and will replace this projection in v3.3.

Table 12: *Projected component-removal estimates on MultiGov-30 ($n = 30$). All numbers are projected from existing message-passing traces using the method above;* none *are full re-runs. Daggers (†) flag projected values.*

Var.	**Description**	$\mathbf{F1}_{\text{token}}$	**GA**	**cF1**	Δ**A0**
A0	Full pipeline (empirical reference)	0.517	0.994	0.756	—
A1†	−SemanticSHACLGate (format-SHACL only)	0.437†	—†	0.219†	−71%†
A2†	−PCST activation (full graph)	0.491†	0.985†	0.738†	−2.4%†
A4†	−Convergent stop (force $R_{\max}$)	0.521†	0.996†	0.759†	+0.4%†
A5†	−Hierarchical aggregation (mean pool)	0.488†	0.994†	0.741†	−2.0%†

† Projected from existing message-passing traces; *not* a full empirical re-run. See projection method above.

Interpretation (projected, with caveats). Under the projection method, the SemanticSHACLGate (A1) is by a wide margin the largest single component effect: removing it is projected to collapse Constrained F1 by 71%. PCST removal (A2) is projected to be a modest 2.4% degradation at $|V| = 32$, but the cost-scaling justification for PCST is Theorem 5.1 (sublinear inference cost), which is not visible at this scale. The convergent-stop (A4) and hierarchical-aggregation (A5) projections are within ±2%, consistent with the small empirical $R_{\max} = 3$ regime where the stopping criterion rarely binds. These projections motivate but do not establish the underlying claims; the empirical re-runs in v3.3 are the binding evidence.

B. Reproducibility

Table 13: *Reproducibility specification.*

Component	**Specification**
Cloud region	AWS `eu-central-1` (Frankfurt, EU data sovereignty)
Reasoning LLM	Claude Sonnet 4.6 via Bedrock (`eu.anthropic.claude-sonnet-4-6`)
Embeddings	`sentence-transformers/all-MiniLM-L6-v2` (384-dim, CPU)
Observer backend	Ollama, Qwen2.5-0.5B (local, deterministic decoding)
Random seeds	42 (Python, NumPy, PyTorch where applicable)
PCST solver	`pcst_fast` (Goemans–Williamson primal-dual)
Activation budget	$K_{\max} = 10$ (static) or AdaptiveActivation tier output
Message-passing rounds	$R_{\max} = 3$
Convergence threshold	$\tau_{\text{conv}} = 0.85$, $\epsilon_{\text{delta}} = 0.02$
Cost budget	$C_{\max} = \$0.50$/query
Gate thresholds	$\text{FF}_{\min} = 0.7$, $\text{SB}_{\min} = 0.7$, $\text{XR}_{\min} = 0.6$, $\gamma_{\min} = 0.7$
Online-learning hyperparams	$\eta = 0.05$, $\gamma = 0.95$, $[w_{\min}, w_{\max}] = [0.1, 5.0]$
Adapter-selector cascade	explicit/domain/fuzzy/none with confidence (1.0/0.7/0.5/0.0)
SDK version	`graqle` v0.42+ (PyPI; pin: see SDK repository tag; package was previously published as co
SDK repository	`https://github.com/quantamixsol/graqle`
Research-artefacts repository	`https://github.com/quantamixsol/cognigraph` (MultiGov-30 benchmark, SemanticSH
Benchmarks	MultiGov-30 JSON at `benchmarks/multigov30.json` (research-artefacts repository)
Reproducibility script	`scripts/reproduce_v32.sh` (research-artefacts repository)

Determinism caveats. Bedrock-hosted Sonnet 4.6 is not strictly deterministic at temperature 0 due to backend batching effects. We report mean over 3 independent seeds where empirical results are flagged; CI widths in Table 9 account for this through bootstrap resampling. Projected ablation rows (Table 12, Appendix A.2) use the single seed-42 message trace.

Compute footprint. A full MultiGov-30 run of CogniGraph (Semantic-SHACL) takes approximately 200 minutes wall-clock and costs \$10.64 in Bedrock charges. Single-Agent baseline takes

3.3 minutes and costs \$1.99.

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